

OCC: Software Developer Computing Theory

COT512

Formative Assessment 1 2026



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DECLARATION OF AUTHENTICITY

I Grant Samson (FULL NAME) hereby declare that the contents of this assessment COT512_FA1 is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any **AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.**

Activity	Date

Signature: 

Date: 18/08/2026

Pre-Assessment Agreement

Assessment Preparation: Preparing the Candidate

Name of Candidate	Grant Samson	Date	17/08/2026
		Time	09:00
Name of Assessor	Mr NB Dube	Venue	Campus
How to prepare the candidate	Document Requirements	Agree (tick)	Action Required
Explain to the candidate why you are meeting and the purpose of the assessment.	Assessment Policy Assessment process	✓	
Discuss the assessment plan in detail.	Assessment strategy	✓	
Explain assessment process, show assessment instruments to candidate and describe assessment conditions.	Assessment instruments	✓	
Identify the role-players during assessment.	Assessors Moderator	✓	
Describe the evidence required to be declared competent.	Examples of evidence	✓	
Explain how evidence will be judged.	Mark allocation explained	✓	
Explain to the candidate how to prepare: Give candidate assessment task description.	Assessment task description	✓	
Confirm with the candidate what he/she should bring to the assessment.	Detailed briefing on exact requirements to be given to candidate in writing	✓	
Ensure that candidate understands the procedures of all assessment practices.	Appeals Policy Appeals procedure Assessment Policy Assessment Procedure Moderation Policy Moderation procedure Verification Policy Verification Procedure	✓	
Ask the candidate if he/she foresees any problems or identify any special needs.	List needs	✓	

Agreed Assessment Plan

Candidate's Name:	Grant Samson		
Assessor's Name:	Mr NB Dube		
Module name:	Computing Theory		
Unit Standard/s:	❖ KM-05-KT01: Introduction to programming languages ❖ KM-05-KT02: Programming basics		
Type of Assessment i.e. Formative assignment, Formative test, Formative Practical, Summative etc.	Formative Assessment		
Special Assessment Requirements:	Formative assessment		
Event	Date, time and location	Resources required	Evidence to be generated
Assessments due dates.	24/08/2026	Assessments	Completed documentation
Complete activity upload to Campus Online	24/08/2026	Study guide	Completed Portfolio of Evidence
Submit Electronic Portfolio of Evidence on EPOE4ALL	24/08/2026	CampusOnline	Acknowledgement of receipt from Facilitator
Assessor Roles and Responsibility			
Roles	❖ Assessor ❖ Guide ❖ Feedback Agent ❖ Reviewer		
Responsibilities	❖ Consult candidate re-assessment, assessment process and plan. ❖ Agree assessment process and plan with candidate. ❖ Forward documentation to candidate: plan, guide and assessment instruments. ❖ Assess candidate with the use of different instruments. ❖ Provide feedback on assessment findings. ❖ Support candidate through assessment process. ❖ Source feedback from candidate on assessment process. ❖ Review assessment process and outcome. ❖ Use assessment process as opportunity to transform assessment activities and outcomes.		

Candidate Roles and Responsibility	
Roles	❖Learner ❖Feedback agent ❖Reviewer
Responsibilities	❖Be available for assessment. ❖Be actively involved in the consultative process. ❖Learn from the assessment process. ❖Provide feedback to the assessor in terms of the assessment as learning activity. ❖Provide feedback to the assessor on the efficacy of the assessment process. ❖Review own role and assessor role in the assessment process.
Assessment Instruments	❖Electronic Portfolio of Evidence ❖Questionnaire ❖Report ❖Presentation ❖Reflexive questions ❖Work sample ❖Practical's ❖Group Activity ❖Research activities
Assessment Process	
❖Evaluation of E-POE addressing Essential Embedded Knowledge in unit standards. ❖Evaluation of Research Projects and other evidence addressing specific unit standards. ❖Consultation: assessment plan and assessment activities and instruments. Pre-assessment moderation and interviews conducted at this stage. ❖Observation: feedback on assessment against specific outcomes, critical outcomes in unit standards. ❖Feedback: to candidate regarding sufficiency of evidence and possible interview to gain supplementary evidence. ❖Feedback to candidate regarding assessment findings as well as review process.	
Feedback	Written feedback to be given to all stakeholders at the end of the assessment process, as well as verbal feedback to the candidate during assessment activities.
Recording Process	Process and findings to be recorded and submitted for record keeping purposes as well as moderation and verification.
Review Process	The review process is the responsibility of the assessor and the candidate. Joint reviewing will take place after feedback has been given to the candidate.
Right to appeal	The candidate must be advised of the right to appeal.

Resources Required	Assignments E-POE Assessments O'Reilly Guides
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I confirm that:

- ❖ I have been consulted on and have agreed to the training and assessment process as detailed in the assessment guide.
- ❖ I have been advised of my right to appeal against any assessment that is unfair, unreliable, invalid or impracticable.
- ❖ I have read and understood the appeal procedure.
- ❖ I know that assessments may be moderated or verified by an external party.
- ❖ The purpose of the assessment has been clearly explained to me.
- ❖ The criteria have been discussed with me, and I know I will be assessed against these criteria.
- ❖ I know when and where I will be assessed, and I was given fair notice.
- ❖ I know how the assessment will be done, and any other requirements related to the assessment.

Signed:  _____

Date: 18/08/2026

QUALIFICATION AND ASSESSMENT DETAILS

Programme Title	OCC: Software Developer
Module Title	Computing Theory
Module Code	COT512
NQF Level	5
Assessment	Formative Assessment 1
Hand Out Date	17/08/2026
Hand in Date	24/08/2026
Total Marks	100
Developer	Kobus Mienie
Internal Moderator	Lerato Tsoehlisi
External Moderator	

Student Name and Surname	Grant Samson
Student Number	20261424

INSTRUCTIONS

- Read each question carefully and consider the mark allocation before answering.
- Ensure you answer all questions.
- For final submission, include the Declaration of Authenticity, Pe-Assessment Document, Assessment document, and all the relevant files required to answer the questions in a zip folder.
- Only a .zip will be allowed for zipped folders
- Font style to be used Calibri (Body) 12pt, unless stated otherwise by a question
- Marks will be deducted for missing or incorrectly named documents

ASSESSMENT OUTCOME

- KM-05-KT01: Introduction to programming languages
- KM-05-KT02: Programming basics

QUESTION 1

(100)

Scenario

You have recently been appointed as a Junior Python Developer at Bright Future High School. The school administration currently records student marks manually, which makes it difficult to organize information and calculate annual averages efficiently. To improve the process, the principal has requested a simple menu-driven Python application that will allow teachers to manage student academic records electronically.

The system must store information for multiple students. For each learner, the program should record:

The student's name.

The grade in which the student is enrolled.

The marks obtained for Term 1, Term 2, Term 3, and Term 4.

The application has the following requirements:

1. Allows continuous user input (5)
2. The following menu should be displayed when the application is opened: (5)

```
Welcome to School Calculator
1. Add students
2. Display student marks
3. Calculate student averages
0. Close

What would you like to do? █
```

3. When the user selects 1: (35)
 - Application prompts the user on how many students should be added.
 - The application prompts the user to add a student name
 - The application prompts the user to enter a student grade
 - The application prompts the user to enter 4 marks (1 mark per term)
 - The application does this for the number of students to be added.
 - Information entered should be stored in a list or dictionary

4. When the user selects 2, on the main menu, all student information should be displayed as follows: (20)

```

Student Name:
=====
Piet van tonder

Student Grade:
=====
10

Student Marks:
=====
Term 1: 50%
Term 2: 60%
Term 3: 70%
Term 4: 75%

Student Name:
=====
Josh Mokoena

Student Grade:
=====
12

Student Marks:
=====
Term 1: 60%
Term 2: 80%
Term 3: 90%
Term 4: 72%

```

5. When the user selects option 3, in the main menu: (20)
- The average marks should be calculated for each student
 - Output should be displayed as follows:

```

Student Average Marks
=====

Student Name: Piet van tonder      Average: 63.75
Student Name: Josh Mokoena        Average: 75.5

```

6. When the user selects option 0, on the main menu the application should close. (5)
7. The application also has the following additional requirements: (10)
 - If a user types an option, not displayed in the menu, a message should be displayed to inform the user of their mistake.
 - The code must contain comments

End of Assessment

MARKING RUBRIC

Question	Task	Marks Allocated	Marks Awarded
1	Application allows continuous user input	5	
2	Main menu correctly displayed (Correct menu options, numbering and formatting)	5	
3	User is prompted for the number of students	3	
	Student names are captured correctly	5	
	Student grades are captured correctly	5	
	Four term marks are captured for each student	10	
	Repetition is performed for all students entered	7	
	Student information is stored in a list or dictionary	5	
4	Option 2 is implemented correctly	4	
	All student information is retrieved from the data structure	8	
	Information is displayed clearly and neatly	8	
5	Average formula is implemented correctly	8	
	Average is calculated for every student	6	
	Results are displayed correctly and clearly	6	
6	Exit option implemented correctly	3	
	Application closes without errors	2	
7	Invalid menu choices are handled correctly	5	
	Appropriate comments are included in the code	5	
Total:		100	

MARK ALLOCATION

Task	Description	Marks
Question 1	Student Performance Management Application	100

LEARNING OUTCOMES INTEGRATED QUESTIONS CROSS-MAPPING GRID

LO No.	Learning Outcome Description	Question	Marks
KM-05-KT01	Introduction to programming languages	1	100
KM-05-KT02	Programming basics		
Total			100